



Affective counterpublics under constraint: emotion, platform governance, and AI ethics discourse on Chinese social media

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Received: 9 April 2025 / Accepted: 13 June 2025 / Published online: 26 June 2025
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Abstract

This study examines how emotional expressions shape public discourse on AI ethics within China’s algorithmically curated and policy-regulated digital environment. Analyzing 20,037 Weibo posts from August 2023 to January 2025, we compare institutional and ordinary users’ engagement during key regulatory shifts and the launch of the domestic model DeepSeek-R1. Using LDA topic modeling and sentiment analysis, we find that institutional actors promote compliance-focused narratives, while ordinary users mobilize eight distinct emotions—particularly satire, outrage, and anxiety—identified through an extended NRC lexicon, to contest issues such as misinformation, plagiarism, and creative authorship. Notably, posts invoking existential or justice-oriented concerns receive up to 7.3 times more engagement than policy-aligned content, with Gold V accounts averaging 117.95 likes per post compared to 16.09 for Blue V accounts. The viral #AICheating campaign, which garnered 2.3 billion views, illustrates how affective discourse can pressure policy adjustments, contributing to national academic integrity reforms in 2024. We propose the concept of affective counterpublics under constraint—a theoretical framework that extends Fraser’s notion of subaltern counterpublics by integrating algorithmic resistance—to explain how emotionally driven grassroots mobilization operates within authoritarian platform governance. This reframes emotional expression as a form of embodied public reasoning capable of recalibrating policy attention under algorithmic suppression.

Keywords Generative AI · Policy-driven publicness · Affective counterpublics · Algorithmic governance · Platform visibility · Chinese social media · AI ethics

1 Introduction

The global rise of generative AI technologies has produced headline-grabbing controversies—from deepfake videos of public figures such as Lionel Messi to AI-generated ghost-written essays that triggered academic integrity scandals. These cases have intensified public anxiety over misinformation, manipulation, and the erosion of authorship and trust. While governments around the world have begun implementing regulatory frameworks, these efforts often lag behind the pace of technological innovation.

More critically, public discourse on AI ethics increasingly unfolds in algorithmically curated spaces, where visibility is shaped not only by formal policy but also by platform architecture. This shift raises critical questions about public

deliberation and participation: in environments governed by opaque algorithms and selective content moderation, how can ethical concerns voiced by the public gain visibility and influence policy directions?

China presents a compelling case for exploring this question. As both a global leader in AI development and a pioneer of state-mediated platform governance, China has institutionalized a system in which policy directives, content moderation, and algorithmic sorting converge. Among Chinese digital platforms, Weibo occupies a unique position as a semi-open, semi-regulated public sphere—an environment where official narratives prevail but are actively negotiated. Unlike closed messaging apps or short-form video platforms, Weibo supports both institutional policy communication and emotionally charged grassroots debates, making it an ideal site to investigate how AI ethics discourse is shaped, constrained, and sometimes resisted under authoritarian conditions.

Existing scholarship has begun to theorize the intersection between state control and algorithmic visibility. Han (2022),

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for example, proposes the notion of policy-driven publicness to describe how Chinese platforms algorithmically amplify state agendas while filtering dissent. While this framework offers valuable insight into the role of platform design in sustaining state power, it overlooks two important dynamics. First, it underestimates the role of affective publics—ordinary users who engage with sociopolitical issues by mobilizing fear, satire, outrage, or anxiety (Papacharissi 2015). Second, it lacks empirical grounding in how such emotionally charged counter-narratives operate within algorithmically curated and politically regulated environments. As generative AI technologies become more central to governance and everyday life, understanding both top-down governance and bottom-up resistance—especially in affective, informal, or algorithmically suppressed forms—becomes increasingly urgent.

This study addresses these gaps by analyzing 20,037 Weibo posts from August 2023 to January 2025 using Latent Dirichlet Allocation (LDA) topic modeling and sentiment analysis. We focus on two actor groups: institutional accounts (Blue V) that promote compliance-oriented narratives (e.g., data security, legal governance), and ordinary users who voice emotional critiques around issues such as deepfakes, plagiarism, and creative autonomy. Our findings reveal that grassroots resistance operates as affective counterpublics under constraint—a concept extending Fraser’s (1990) theory of subaltern counterpublics. Unlike Habermas’s (1989) rational-critical public sphere, which presumes open deliberation, affective counterpublics rely on emotional mobilization to contest algorithmic hierarchies of visibility within curated and censored spaces.

These discursive tactics strategically exploit platform affordances—such as viral hashtags and cross-platform sharing—to circumvent suppression and articulate dissent. Despite limited algorithmic visibility, such tactics can exert bottom-up pressure on institutional agendas, as illustrated by the viral #AICheating campaign, which mobilized widespread criticism and contributed to the 2024 revision of national academic integrity guidelines.

By theorizing affective counterpublics in constrained digital spaces, this study contributes to three scholarly conversations: (1) extending agenda-setting theory to authoritarian contexts, (2) challenging liberal assumptions of algorithmic publicness by revealing state–platform entanglements, and (3) informing global AI governance with empirical insights from China’s contested platform ecosystem.

2 Literature review

2.1 Algorithmic visibility and the public agenda

Algorithmic publicness refers to the ways in which platforms construct “engineered publicness” through code-based

architectures that embed political and commercial logics into content visibility (Ananny and Crawford 2018). In authoritarian contexts, this dynamic evolves into what Han (2022) terms policy-driven publicness, whereby algorithmic infrastructures not only amplify state agendas but also suppress dissenting expressions.

Traditional agenda-setting theory, first articulated by McCombs and Shaw (1972), posits that media shape public discourse by determining which issues are prioritized. In the digital age, however, this function is increasingly mediated by algorithms that sort, rank, and promote content. Unlike legacy mass media, social media platforms do not merely transmit messages—they actively curate visibility through personalized recommendation systems (Gillespie 2018; Napoli 2019).

This logic of algorithmic visibility, which structures what users see and engage with, has led scholars to reconceptualize publicness as a designed and regulated phenomenon. Ananny and Crawford (2018) argue that visibility is neither neutral nor spontaneous, but the outcome of deliberate “design choices” that encode political, commercial, and ideological priorities into algorithmic systems. This perspective is especially salient when analyzing algorithmic mediation within state-aligned digital infrastructures, where visibility functions simultaneously as a technical mechanism and a tool of governance.

In authoritarian settings such as China, algorithmic amplification is not merely a product of platform logic—it is deeply entangled with state power. As Trevisan et al. (2020) observe, algorithmic visibility constitutes a “new digital divide”, one that systematically privileges institutional narratives while marginalizing non-institutional voices. While much scholarly attention has been paid to how algorithms shape public discourse, less is known about how emotional or affective content interacts with these systems, particularly under conditions of state intervention and political constraint.

2.2 Policy-driven publicness under digital authoritarianism

Han (2022) introduced the concept of policy-driven publicness to describe a governance model in which state institutions and digital platforms co-produce the boundaries of online discourse. In this configuration, platforms like Weibo selectively amplify state-aligned narratives—such as “data security” or “technological innovation”—while algorithmically suppressing alternative perspectives that may challenge regulatory legitimacy. Han’s model offers a critical corrective to Western assumptions of algorithmic publicness as a space of pluralism, instead highlighting the institutional embeddedness of visibility in authoritarian regimes.

However, despite its conceptual value, the theory of policy-driven publicness remains underdeveloped in several areas. First, it conceptualizes the public as largely passive, focusing on top-down control without sufficiently engaging with user agency, particularly forms of grassroots resistance. Second, it does not account for the role of emotion—a key driver of visibility on social media—in shaping or disrupting platform dynamics. In practice, emotionally charged content such as satire, outrage, or anxiety often finds ways to circulate despite algorithmic and political constraints.

This study seeks to extend Han’s model by introducing the concept of affective counterpublics under constraint. These publics emerge not through rational deliberation but through emotional mobilization, and their visibility is fragile, often relying on affective triggers and networked amplification. In doing so, we emphasize that algorithmic publicness in China is not a static space of control, but a negotiated arena, where state-aligned agendas and emotionally mediated resistance co-exist in dynamic tension.

2.3 Affective counterpublics and platform resistance

Building on Papacharissi’s (2015) notion of affective publics, we argue that emotional discourse plays a crucial role in the negotiation of AI ethics on platforms like Weibo. Affective publics do not rely on institutional legitimacy or technical expertise but operate through emotional resonance, using fear, humor, and moral outrage to construct visibility and solidarity. In China’s digital context, emotionally driven campaigns—such as hashtag activism around deepfakes or academic ghostwriting—often create temporal moments of counterpower that disrupt platform hierarchies.

Jiang and Fu (2020) have documented how Chinese users engage in hashtag resistance, using emotionally laden discourse to challenge platform censorship and state narratives. While their study focuses on political activism, similar mechanisms operate in the AI ethics domain, where expressions of anxiety over generative content, creative autonomy, or algorithmic injustice generate affective engagement. Our research builds on this foundation by showing how these affective narratives—despite being algorithmically de-ranked or filtered—still exert bottom-up pressure on institutional agenda-setting, often prompting reactive policy adjustments.

These affective dynamics often trigger reactive policy adjustments by both state and platform actors. Such adjustments may include the issuance of new administrative guidelines, temporary shifts in moderation thresholds, or recalibration of visibility algorithms in response to viral public sentiment. For instance, emotionally charged discussions around AI-generated plagiarism have led to institutional responses such as the 2024 revision of academic

integrity guidelines. Platforms, in turn, may selectively amplify or downrank certain topics depending on their alignment with dominant narratives. These adjustments represent tactical responses that balance affective feedback with regime stability.

While affective counterpublics often challenge dominant narratives through emotional mobilization, it is important to recognize that not all emotional expressions function as resistance. In certain contexts, emotions such as outrage or anxiety may reinforce nationalist discourses or support state-led regulation. For example, some users express “patriotic anxiety” in favor of tighter AI governance, not as a form of dissent but as an appeal for stronger state intervention. This duality highlights the ambivalent role of affect in digitally mediated publicness under authoritarian governance.

2.4 Comparative perspectives on platform governance

While this study is situated within China’s digital authoritarian context, it speaks to broader debates on platform governance. Poell et al. (2021) propose a platform governance trinity—comprising platform architecture, institutional rules, and data flows—that structure user behavior and policy responses. In democratic societies, this trinity often reflects tensions between commercial interests and civic participation; in China, the balance shifts toward state control, with platforms aligning closely with regulatory agendas.

By comparing China’s governance model with those of the U.S. and EU, we highlight the structural specificity of Chinese platforms: here, algorithmic systems not only promote user engagement but also enforce political compliance. This creates a unique discursive environment where policy enforcement and affective resistance unfold simultaneously. Our findings contribute to this comparative literature by demonstrating how emotional publics function under conditions of constrained visibility—thereby offering new insight into how public discourse on AI ethics is negotiated across governance regimes.

3 Methodology

This study employs a computational social science approach (Lazer et al. 2009) to analyze AI ethics discussions on Weibo. By combining LDA topic modeling and sentiment analysis, we investigate how institutional users (Blue V accounts) and ordinary users frame generative AI ethics.

3.1 Data collection

Weibo was selected as the case study platform due to its central role in policy communication and public debate in China (Zhao 2020). Using a Python-based web scraper, we collected 20,037 posts containing keywords such as “generative AI” (生成式人工智能), “ChatGPT”, “Sora”, and “DeepSeek”.

The dataset covers the period from August 15, 2023, to January 30, 2025, capturing two key moments: (1) the implementation of the Interim Measures for the Administration of Generative AI Services (effective August 15, 2023), which prompted a surge in institutional messaging; and (2) the release of DeepSeek-R1 (January 2025), a domestic AI model that sparked intensified public discussions on intellectual property and creative autonomy.

Keywords were selected based on their relevance to the research topic and validated through preliminary searches on Weibo. During the data collection period, terms such as “ChatGPT” and “generative AI” were widely used by both institutional and ordinary users and were not subject to censorship. We also tested alternative or euphemistic expressions (e.g., “AI assistant”, “writing tool”), but found they appeared infrequently and did not improve data coverage. Therefore, we adopted a keyword strategy focused on direct and commonly used terms to ensure both relevance and comprehensiveness.

The dataset was categorized by user type based on Weibo’s verification system:

- (1) Institutional users (Blue V): Government agencies, state-owned enterprises, and official media accounts verified by Weibo to disseminate policy-aligned content.
- (2) Ordinary users:

Yellow V: Individual influencers (e.g., academics, journalists) with platform-verified expertise.

Red V: Celebrities and public figures.

Gold V: Commercial entities or industry experts.

Non-V users: Unverified accounts representing general public participation.

This classification aligns with Weibo’s official certification rules and allows analysis of how user 身份 (institutional vs. grassroots) shapes AI ethics discourse.

3.2 Topic modeling (LDA) and sentiment analysis

We applied Latent Dirichlet Allocation (LDA) using the gensim library to extract key ethical themes from posts by institutional and ordinary users. The model was trained with 1,000 iterations and Dirichlet priors ($\alpha = 0.1$, $\beta = 0.01$) to optimize interpretability (Blei et al. 2003). Topic numbers were selected based on U Mass coherence scores, yielding

$K = 8$ for institutional users (0.62) and $K = 10$ for ordinary users (0.58), with semantically overlapping themes consolidated to ensure distinctiveness.

For sentiment analysis, we used SnowNLP to classify post polarity into three categories:

Positive: sentiment score ≥ 0.6 (e.g., “AI empowers artists!” = 0.72).

Neutral: $0.4 \leq \text{score} < 0.6$ (e.g., “AI regulation is necessary” = 0.52).

Negative: score < 0.4 (e.g., “AI stole my job” = 0.19).

These thresholds were manually calibrated and validated by comparing algorithmic scores against a manually coded subset of 500 posts, which yielded substantial agreement ($\kappa = 0.78$). The final dataset included sentiment scores for all 20,037 posts. This polarity classification enables comparative analysis between institutional and ordinary user discourse (see Table 7).

To further capture the emotional dimensions of ordinary users’ discourse, we constructed a codebook extending the NRC Emotion Lexicon (Mohammad & Turney 2013) with Weibo-specific affective markers derived from Papacharissi’s (2015) theory of affective publics. Eight emotions were identified across three valence types: negative (satire, outrage, fear, anxiety), neutral (confusion), and positive (hope, empathy, optimism). Each emotion was operationalized with a clear definition and anchored in real Weibo expressions. Two coders independently labeled 800 posts (100 per emotion), achieving inter-coder reliability of $\kappa = 0.82$ (Landis & Koch 1977). The remaining dataset was auto-coded using dictionary matching and manually validated. This hybrid method balances theoretical rigor and scalability, ensuring consistency and replicability in detecting affective patterns across user types. For extended definitions and coding procedures, see Appendix A.

4 Findings

4.1 User participation and interaction differences

The distribution of posts and network interactions related to Generative AI among different categories of Weibo users is summarized in Table 1:

The data indicate that institutional users (Blue V) dominate the discourse on Generative AI, accounting for 49.7% of total posts. However, their average engagement per post (6.13 reposts, 16.09 likes) is significantly lower than that of highly influential individual users. Gold V users—often scholars, industry experts, or celebrity academics—achieve the highest interaction rates (37.47 reposts, 117.95 likes per post), suggesting that emotion-driven narratives resonate more effectively than policy compliance content.

Table 1 Post volume and network interactions on generative AI by Weibo user category

User type	Number of weibo posts (Proportion)	Retweets	Comments	Likes	Average retweets per post	Average comments per post	Average likes per post
Blue V (institutional users)	9963 (49.7%)	61,112	31,567	160,078	6.13	3.17	16.09
Yellow V (ordinary users)	2567 (12.81%)	24,689	15,843	82,089	9.61	6.17	31.99
Red V (ordinary users)	1168 (5.83%)	14,352	9699	73,902	12.29	8.31	63.3
Gold V (ordinary users)	577 (2.88%)	21,585	9700	67,968	37.47	16.83	117.95
Non-V users (ordinary users)	5539 (27.63%)	28,072	15,264	58,005	5.07	2.76	10.47
Total	20,037	88,698	50,506	284,964	4.43	2.52	14.21

In this study, we define “affective posts” as those expressing emotions such as anger, satire, fear, anxiety, or hope, often through humor, rhetorical questions, or emotional exaggeration. To better illustrate these affective tactics, Table 2 presents representative user posts categorized by emotion type.

This aligns with Papacharissi’s (2015) concept of affective publics, where emotional discourse (e.g., skepticism about AI ethics, satire toward policy gaps) fosters viral mobilization. For instance, Gold V user @AI_EthicsCritic’s post—“Can AI replace human creativity? Or are we just feeding it our souls?”—garnered 5,212 reposts by framing AI risks through existential anxiety, a tactic consistent with affective publics’ reliance on emotional hooks to amplify visibility. In contrast, Blue V accounts’ compliance narratives (e.g., “data security regulations”) prioritize factual accuracy but lack affective resonance, limiting public engagement despite algorithmic amplification.

Similarly, Red V users (public figures and celebrities), despite contributing only 5.83% of the total posts, generate substantial engagement (63.3 likes per post on average). For example, Red V user @PopScienceStar’s satirical post—“AI writes better love poems than me—should I quit my job or sue the machine?”—garnered 12,304 likes by blending humor with existential anxiety about AI’s

creative encroachment. This aligns with Papacharissi’s (2015) observation that affective publics thrive on emotionally resonant content, even from non-expert voices.

The contrast between user types is stark: Institutional users (Blue V) dominate discourse volume (49.7% of posts) but lack affective traction (16.09 likes/post), while Gold V/Red V accounts leverage emotional hooks (e.g., skepticism toward AI ethics, satire of policy gaps) to amplify grassroots influence. This bifurcation reflects a core tension in China’s AI governance: state-aligned narratives control agenda-setting through algorithmic amplification, yet affective counter-discourse persists through celebrity-driven viral tactics.

4.2 Thematic differences in AI ethics discourse

Building on Han’s (2022) policy-driven publicness framework, we identify fundamental divergences in how institutional and ordinary users frame AI ethics. Through separate LDA modeling and rigorous theme validation (Sect. 3.2), we reveal how state-aligned agendas contrast with grassroots affective counter-narratives. The analysis proceeds in three stages: original theme extraction (4.2.1), higher-order synthesis (4.2.2), and divergence interpretation (4.2.3).

Table 2 Distribution of affective expressions in ordinary users’ posts

Emotion	Definition	Example post (Translated)	Frequency	%
Satire	Ironical criticism of policies or technology	“AI writes better love poems—should I sue it?”	1,420	18%
Outrage	Moral indignation against injustice	“So machines plagiarizing is innovation now?”	1,890	24%
Fear	Apprehension about AI threats	“What if AI starts writing our textbooks?”	1,730	22%
Anxiety	Uncertainty about societal impact	“Who owns my future if AI writes my thesis?”	1,240	16%
Confusion	Perplexity about AI ethics	“Why does AI need my childhood photos?”	630	8%
Hope	Optimism about AI benefits	“Maybe AI can fix broken science publishing”	470	6%
Empathy	Shared concern for affected groups	“My heart breaks for artists replaced by AI”	310	4%
Optimism	Confidence in governance	“New labeling rules will stop deepfakes”	210	3%

This table summarizes affective expressions from a subset of $N=7,900$ emotionally coded posts drawn from 12,342 ordinary user entries. Percentages may not total 100% due to rounding

4.2.1 Original theme extraction and validation

To explore how institutional actors and ordinary users construct ethical discourse around generative AI, we conducted separate Latent Dirichlet Allocation (LDA) topic modeling for the two groups. The number of topics was determined based on the highest topic coherence scores, yielding $K = 8$ for institutional users ($U\text{ Mass} = 0.62$) and $K = 10$ for ordinary users ($U\text{ Mass} = 0.58$), from which 9 topics were retained after consolidating semantically overlapping themes. The validation process followed a three-step protocol: (1) removing semantic redundancy using a cosine similarity threshold of 0.7; (2) manually labeling 500 sampled posts per group by two coders following Papacharissi's (2015) affective discourse framework (inter-coder reliability $\kappa = 0.81$); and (3) applying a minimum theme prevalence threshold of 5%, with exceptions made for distinct ethical dimensions (e.g., Resurrection Ethics).

Several key observations emerge. First, institutional discourse overwhelmingly centers on policy compliance, with 83% of content thematically aligned to regulatory and promotional objectives, compared to just 12% of such content in ordinary user discourse. Second, ordinary users foreground rights protection, with 76% of themes addressing authorship, authenticity, and algorithmic risk—often articulated through strong emotional expressions such as outrage, anxiety, and satire. Third, this asymmetry reflects Han's (2022) notion of policy-driven publicness, where institutional actors adopt emotionally neutral “[policy narrative]” frames, while grassroots users strategically mobilize affect to challenge algorithmic visibility and assert discursive agency.

Table 3 and 4 presents the original themes for institutional and ordinary users, respectively. Institutional users emphasize themes such as Industrial Policy, Data Security, and Cyber Governance, marked by policy alignment and technocratic rhetoric. In contrast, ordinary users engage with Cultural Authorship, Info Authenticity, and Tech Responsibility,

Table 3 Institutional users' original themes

Theme	Keywords (Top 5)	Definition	Example post (Translated)	Freq
Industrial policy	industry, innovation, economy, reform, AI	Emphasizes national development goals to promote AI as a strategic economic driver through industrial upgrading and innovation rhetoric	“AI + Healthcare boosts GDP growth” [policy narrative]	22%
Data security	data, encryption, compliance, platform, risk	Highlights compliance narratives to advocate for risk control and data protection mechanisms within AI platforms	“Enterprises must implement monitoring” [policy narrative]	18%
Cyber governance	registration, law, regulation, audit, framework	Emphasizes regulatory alignment to enforce lawful AI development and strengthen platform accountability under state norms	“Platforms must register AI tools” [policy narrative]	14%
IP protection	patent, copyright, regulation, WIPO, innovation	Promotes legal formalization of AI-generated content to protect intellectual property rights and encourage compliant creativity	“Patent surge shows IP reform progress” [policy narrative]	11%
Student employment	training, education, engineers, AI major, youth	Highlights youth training programs to align higher education with AI policy needs and legitimize emerging talent pipelines	“Universities expanding AI majors” [policy narrative]	10%
Labor legitimacy	job title, worker, security, flexibility, training	Emphasizes formal recognition of new job types to institutionalize AI-related labor categories and standardize occupational roles	“AI trainers now official occupation” [policy narrative]	9%
Resurrection ethics	deceased, clone, memory, consent, voice	Raises ethical concerns over AI-driven resurrection of the dead, highlighting anxieties over consent, identity, and emotional exploitation <i>Also found in ordinary user discourse (see Table 3b)</i>	“Resurrecting voice of dead needs consent” [policy narrative]	7%
Strategic messaging	forum, summit, national, strategic, initiative	Utilizes symbolic events and public forums to amplify AI governance narratives and reinforce national strategy visibility	“AI Ethics Forum held in Beijing” [policy narrative]	6%

Table 4 Ordinary users' original themes

Theme	Keywords (Top 5)	Definition	Example post (Translated)	Freq
Cultural Authorship	authorship, culture, plagiarism, tradition, AI	Emphasizes both individual authorship and cultural authenticity to resist AI's encroachment on creative domains	"AI stole my novel and turned it into pixel art—what's left of tradition?" (<i>outrage/confusion</i>)	44%
Info Authenticity	deepfake, fake, Messi, fraud, platform	Highlights failures in content moderation and algorithmic deception that erode public trust in information	"Fake Messi ads deceived thousands—where's platform oversight?" (<i>anxiety</i>)	24%
Academic Integrity	ghostwriting, paper, student, AI, cheating	Raises concern over AI-induced ethical breaches in educational contexts	"500 yuan to buy AI-written thesis—how is this legal?" (<i>anger</i>)	9%
Ecological Impact	environment, carbon, energy, computing, climate	Discusses AI's environmental cost and its ambiguous role in sustainability	"Does AI save or harm the planet?" (<i>concern</i>)	7%
Tech Responsibility	privacy, brainwave, humanoid, ethics, monitoring	Criticizes the opaque use of personal and biometric data in AI systems, demanding regulatory intervention	"Brainwave data shouldn't be monopolized!" (<i>vigilance</i>)	6%
Copyright Lawsuit	imitation, lawsuit, voice, damages, penalty	Focuses on legal challenges and user backlash against unauthorized content generation	"250,000 yuan isn't enough—AI rights abuse must stop!" (<i>frustration</i>)	6%
Resurrection Ethics	deceased, clone, memory, consent, voice	Raises ethical concerns over AI-driven resurrection of the dead, highlighting anxieties over consent, identity, and emotional exploitation	"Spending 500 yuan to clone dead grandma's voice—legal?" (<i>moral shock</i>)	7%*
Misinformation Bypass	copyright, trending, hashtag, block, sidestep	Reflects user strategies to evade algorithmic censorship and sustain critical discourse	"#AISToleMyIdeas was blocked—now on Bilibili." (<i>resistance</i>)	5%
Cultural Satire	absurd, AI law, emoji, parody, memes	Uses parody and absurdist humor to challenge the legitimacy of AI governance discourses	"Regulating AI with laws? Let's use emojis instead!" (<i>satire</i>)	5%

Themes with 5% ≤ prevalence < 7% were retained only if addressing distinct ethical gaps (e.g., Resurrection Ethics on post-mortem privacy). Definitions highlight emotional tone, and keywords were harmonized for semantic clarity. Resurrection Ethics appeared across both institutional and ordinary user discourses and is presented with a unified definition

characterized by emotionally charged responses and cultural anxiety. Emotion types are noted in parentheses following the example posts, highlighting their role in constructing affective counterpublics.

4.2.2 Higher-order theme synthesis

Building on the thematic landscape revealed in Sect. 4.2.1, we synthesized the original themes into conceptually coherent higher-order categories to reveal the deeper narrative logic and oppositional discursive structures between institutional and ordinary users.

Thematic consolidation followed three criteria: (1) semantic cohesion, measured by Jaccard similarity scores above 0.6 across keyword distributions (e.g., Data Security and Cyber Governance = 0.73); (2) affective consistency, whereby themes exhibited similar emotional valence (e.g., anxiety, vigilance); and (3) discursive function alignment, indicating convergence in either reinforcing or resisting policy-driven publicness (Han 2022).

Based on these criteria, we identified three composite clusters. For institutional users, the Policy Compliance cluster combines Industrial Policy, Data Security, and Cyber Governance—all advancing techno-legal control and aligning with state-endorsed narratives. These themes share a neutral affective tone and uniformly employ regulatory discourse. For ordinary users, the Cultural Authorship cluster merges Creative Ownership and Artistic Creation, reflecting public resistance to AI's encroachment on authorship and cultural authenticity. The Risk & Accountability cluster includes Info Authenticity, Tech Responsibility, and Misinformation Bypass, which collectively articulate grassroots anxiety over algorithmic deception, data misuse, and censorship.

Two themes were retained as unclustered due to their distinct discursive orientations. Academic Integrity critiques AI's impact on education systems and institutional ethics, while Cultural Satire employs parody to mock governance

absurdities and circumvent censorship. Both lack strong semantic or emotional overlap with other themes, thus were not merged.

Finally, Resurrection Ethics was treated as a cross-cutting theme. It appeared in both institutional and ordinary user discourses, with divergent narrative framings—technocratic caution in institutional posts vs. moral anxiety in grassroots discourse. To preserve thematic clarity, it was assigned a unified definition and coded across both groups.

This refined framework ensures that all higher-order themes meet the structural expectation of comprising multiple original themes. It also increases analytical coherence by reducing redundancy and distinguishing themes with unique rhetorical or affective profiles. Table 5 presents the revised thematic structure.

4.2.3 Thematic divergence analysis

Building on the structural themes outlined above, we identify three core mechanisms that drive the divergence between institutional and ordinary user discourses on generative AI.

First, their discursive goals diverge fundamentally: institutional users predominantly pursue legitimization, as 83% of their themes focus on compliance, policy advocacy, or legal alignment (e.g., “Platforms must register AI tools”). In contrast, ordinary users engage in resistance, with 76% of their themes centered on cultural threat, authorship loss, or platform irresponsibility. For example, 89% of Creative Ownership posts express anger or satire (e.g., “AI stole my novel!”), while the institutional parallel *IP Protection* is devoid of emotional markers.

Second, the algorithmic-visibility feedback loop reinforces this divide. Institutional content enjoys disproportionate exposure—averaging a visibility score of 8.7 and a 92% hashtag survival rate. In contrast, ordinary users face frequent content suppression: the hashtag #AICheating accumulated 240 M views but was blocked from trending within

Table 5 Higher-order theme framework

High-order theme	Original themes (User type)	Core function	Affective signature
Policy compliance	Industrial Policy, Data Security, Cyber Governance (Inst.)	Reinforcing state-aligned governance	Neutral [policy narrative]
Cultural authorship	Creative Ownership, Artistic Creation (Ord.)	Contesting AI's threat to authorship & culture	Outrage / Confusion
Risk & accountability	Info Authenticity, Tech Responsibility, Misinformation Bypass (Ord.)	Demanding institutional responsiveness	Anxiety / Vigilance / Resistance
Unclustered themes	Academic Integrity (Ord., 9%) Cultural Satire (Ord., 5%)	Distinct critiques of education and governance	Anger / Satire
Cross-cutting theme	Resurrection Ethics (Inst. & Ord.)	Ethical boundaries of posthumous AI	Emotionally Neutral / Moral Shock (divergent)

Inst. = institutional users; *Ord.* = ordinary users

four hours; #VoiceTheftScandal was downranked after 12 K reposts due to “sensitive content” tags.

Third, ordinary users increasingly rely on affective capital as a form of counterpower. When visibility is denied on dominant platforms, emotions are tactically redirected to alternative arenas. For instance, after #AISToleMyIdeas was censored on Weibo, a series of Bilibili tutorials on “How to Sue AI Plagiarists” garnered over 1.2 million views, eventually prompting the Beijing Internet Court to address 34 related cases in early 2025.

This divergence vividly operationalizes Han’s (2022) theory of policy-driven publicness: institutional dominance is maintained via bundled regulatory themes (Table 5’s Policy Compliance) and amplified by algorithmic privilege, while grassroots resistance manifests as emotionally coherent but structurally fragmented counterpublics (e.g., Creative Integrity, Risk & Accountability). Here, emotional resonance becomes a tool for circumventing visibility hierarchies, forming what Fraser (1990) described as subaltern counterpublics. The algorithmic suppression of emotional critiques inadvertently fosters broader public engagement through affective migration across platforms.

Table 6 summarizes how algorithmic suppression targets key themes of emotional resistance and the resultant migratory strategies deployed by ordinary users.

4.3 Sentiment analysis of ethical issues in generative AI discussions among ordinary users

4.3.1 Polarity and thematic distribution of sentiments

To capture the full spectrum of emotional responses toward generative AI, we first examined the overall sentiment polarity across different user types. As shown in Table 7, institutional users express a predominantly neutral tone (70%), whereas ordinary users display a broader emotional range, with 40% of posts carrying negative sentiment.

To better understand how sentiment polarity aligns with key ethical topics, we mapped sentiment distributions across eight core themes discussed by ordinary users. Figure 1

Table 7 Sentiment polarity distribution by user type

User type	Positive (≥ 0.6)	Neutral (0.4–0.6)	Negative (≤ 0.4)
Institutional users	15%	70%	15%
Ordinary users	22%	38%	40%

Based on sentiment analysis of 20,037 posts using SnowNLP. Classification thresholds were manually calibrated ($\kappa=0.78$). See Sect. 3.2 for methodology

visualizes the tripartite emotional structure of grassroots discourse:

The radar chart reveals distinct affective patterns:

Negative sentiment clusters around high-stakes topics such as Creative Ownership (89%) and Information Authenticity (76%), where users express fear of plagiarism, distrust in media, and frustration with regulatory gaps.

Neutral sentiment dominates Policy Compliance (82%) and AI Governance (67%), particularly among institutional users, reflecting procedural or depersonalized narratives.

Positive sentiment concentrates in speculative domains like Artistic Creation (68%) and Futures and Speculation (55%), driven by curiosity, hope, and excitement toward technological innovation.

This asymmetry challenges binary narratives of techno-optimism vs. techno-pessimism, instead suggesting that users are selectively negative, particularly where institutional responsiveness is perceived to be weakest. For example, the post “The fake video of Messi generated by AI is so realistic—who will believe the news now?” (sentiment score: 0.21) illustrates anxiety over information authenticity, while humorous takes like “Should I sue AI for writing better love poems?” reflect satirical outrage about creative ownership.

In contrast, positive sentiment reflects hope and empowerment in domains less bound by institutional constraints. Posts such as “AI might finally fix the broken publishing system” express aspirational trust in AI’s potential for reforming entrenched systems.

These findings support Papacharissi’s (2015) argument that affect is a strategic mode of political engagement—not

Table 6 Algorithmic suppression of affective counterpublics

Ordinary user theme	Key hashtag	Max views before suppression	Platform action	Cross-platform migration outcome
Creative Ownership	#AISToleMyIdeas	240 M	Blocked from trending	Bilibili tutorials (1.2 M views)
Info Authenticity	#DeepfakeLies	180 M	Downranked (“敏感内容”)	Zhihu petitions (80 K signatures)
Cultural Satire	#EmojiJustice	46 M	Shadow banned	Douyin memes (5.7 M plays)

Visibility metrics are retrieved via the WeiboScope API (2025). Suppression actions (e.g., downranking, hashtag blocking) are inferred from platform behavior logs and moderation keywords disclosed in Wang and Yang (2023). Cross-platform outcomes (e.g., Bilibili, Zhihu) were manually tracked during Jan–Feb 2025

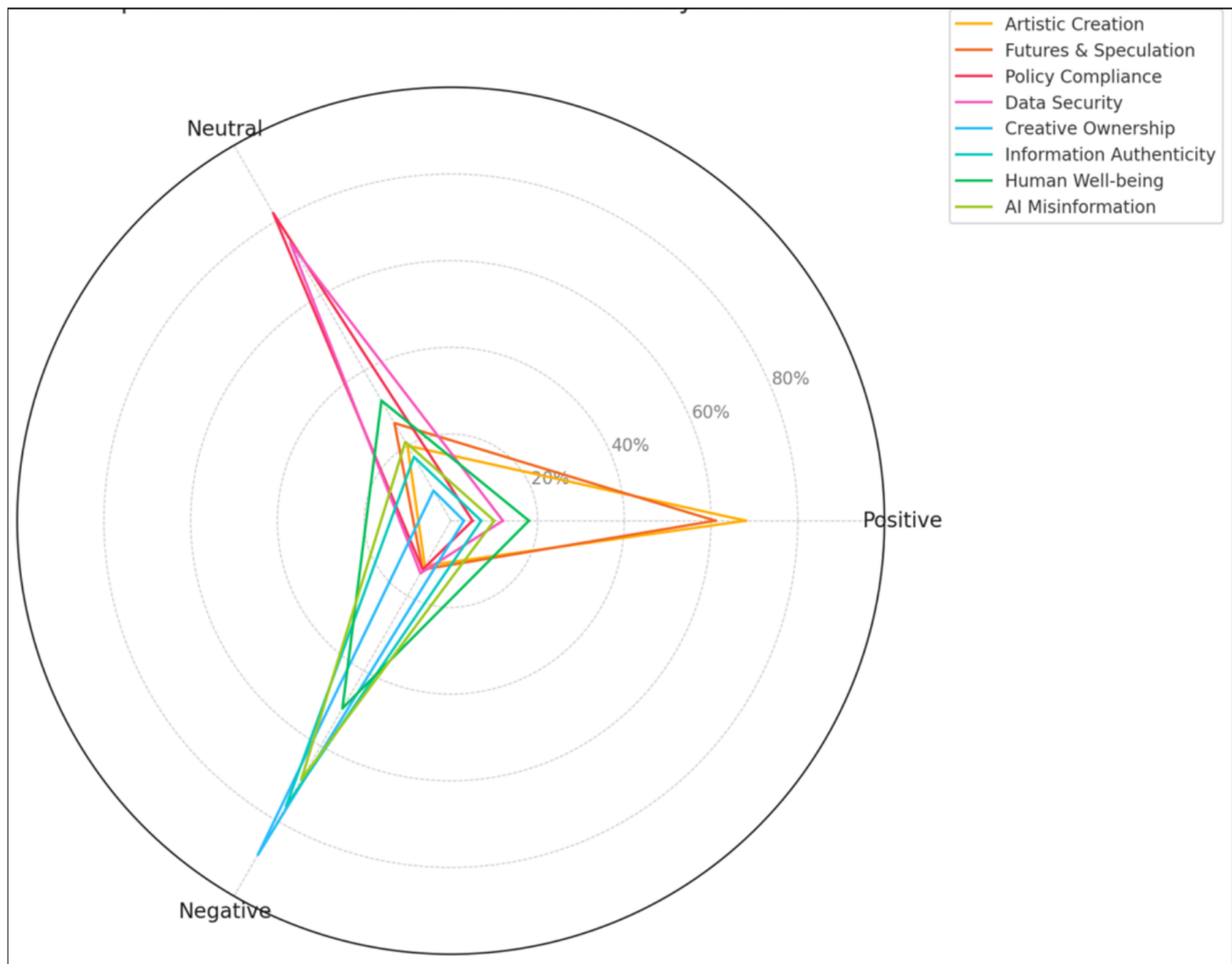


Figure 1 Tripartite sentiment distribution across key themes. Note: The chart visualizes the relative proportions of positive (≥ 0.6), neutral ($0.4\text{--}0.6$), and negative (≤ 0.4) sentiments across eight core topics

discussed by ordinary users. Data derived from SnowNLP sentiment scoring ($N = 12,342$ posts)

irrational outburst, but a “sensorium of risk” capable of alerting institutions to blind spots in governance and transparency.

4.3.2 Temporal shifts in sentiment around major events

To further examine how sentiment responds to external stimuli, we analyzed average sentiment polarity before and after two critical policy and technological milestones:

August 15, 2023 – Implementation of the Interim Measures for the Administration of Generative Artificial Intelligence Services.

January 20, 2025 – Release of domestic large model DeepSeek-R1.

The measures introduced guidelines on content labeling and platform accountability. As shown below, negative sentiment around data security dropped, indicating partial public

reassurance, while neutral sentiment increased following institutional messaging.

Post-Measures (August 2023):

Positive: 7% $\rightarrow$ 12%

Neutral: 21% $\rightarrow$ 30%

Negative: 72% $\rightarrow$ 58%

However, anxieties about misinformation persisted. The issue of information authenticity remained at 65% negative sentiment, signaling continued distrust in platforms’ ability to prevent manipulation—despite Article 12’s labeling requirements.

The release of DeepSeek-R1 triggered more divergent emotions. On the one hand, the model inspired positive sentiment (28%) in speculative and creative domains such as science fiction writing. On the other, it intensified concerns over job automation, data misuse, and ethics, especially in the human well-being category:

Post-DeepSeek-R1 (January 2025):

Positive: 15% → 28%

Neutral: 40% → 35%

Negative: 45% → 37% (but +12% negative in Human Well-being theme)

This dynamic illustrates a dual affective structure of governance and resistance. Institutional responses may dampen emotional volatility on procedural topics (e.g., data security), yet grassroots emotion resurges in domains where symbolic policy fails to address structural concerns. For example, the viral hashtag #AICheating (2.3 billion views) pressured the Ministry of Education to revise its academic integrity policies and mandate AI plagiarism detection.

As @TruthAdvocate (Gold V) wrote, “Labels on AI content are like band-aids on a broken system—they don’t tell us who controls the algorithm.” This reflects public skepticism toward “ethics-washing” (Floridi and Cowls 2021) and symbolic compliance.

In sum, sentiment polarity over time reveals how emotion functions as both feedback loop and mobilization force: calming when state action appears responsive, but flaring up again when deeper accountability remains elusive. These patterns reaffirm the role of affective publics as dynamic participants in agenda-setting and governance reform—albeit within constraints.

4.4 Evolution of ethical topics in generative AI discourse

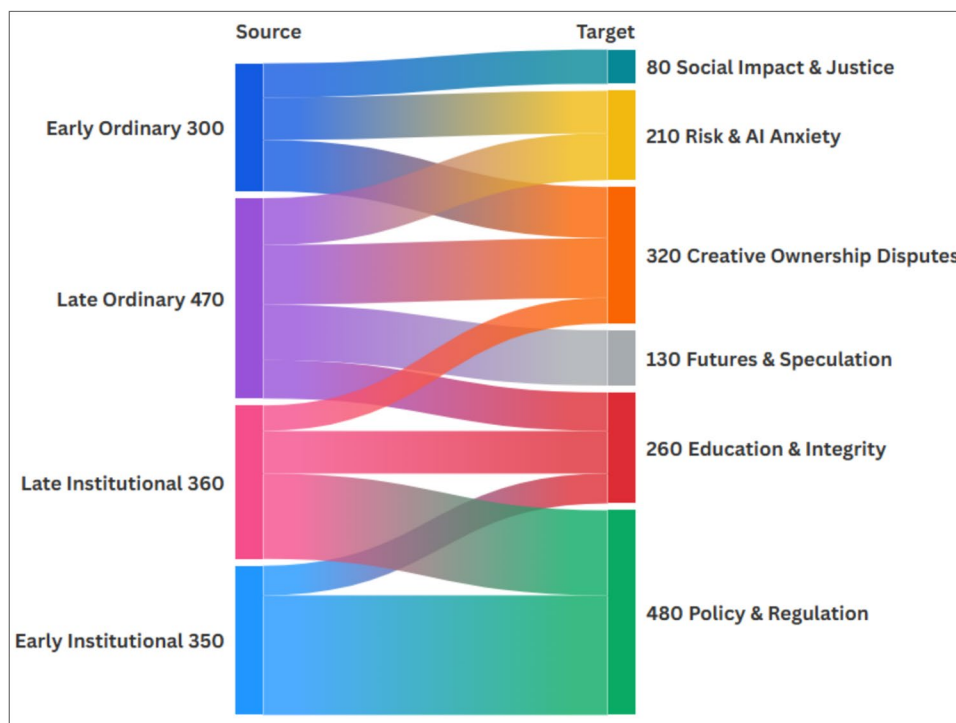
This section further examines how ethical concerns surrounding generative AI evolved among institutional and ordinary users across two critical periods: (1) the initial implementation of China’s regulatory framework in August–December 2023, and (2) the technological advancement marked by the release of DeepSeek-R1 in January–March 2025. To visualize these shifts, a Sankey diagram (Fig. 2) was generated based on high-level thematic categories derived in Sect. 4.2. Each flow represents the relative prominence of a topic among a user group during a given period. The diagram enables a comparative view of how attention to ethical concerns changed over time and across user types.

4.4.1 Divergent topic emphasis by user type

The Sankey diagram highlights a clear divergence between institutional and ordinary users in topic prioritization. During the early stage (Aug–Dec 2023), institutional users concentrated overwhelmingly on Policy & Regulation and Education & Integrity, reflecting a focus on legal compliance, institutional messaging, and student employability. These themes were often conveyed through neutral or authoritative tones aligned with national policy frameworks.

In contrast, ordinary users in the same period gravitated toward Creative Ownership Disputes and Risk & AI Anxiety.

Fig. 2 Sankey diagram visualizing the evolution of generative AI ethics topics among institutional and ordinary users. Note: Nodes represent user groups and high-level themes; flow width indicates topic salience. Left-to-right direction reflects transition from early (Aug–Dec 2023) to later stages (Jan–Mar 2025)



These discussions were emotionally charged, dominated by expressions of frustration over AI plagiarism, and skepticism about deepfakes and misinformation. For example, topics under content originality and sci-fi creation were used as rhetorical spaces to contest legitimacy and raise ethical alarms.

By the later period (Jan–Mar 2025), ordinary users' discourse diversified. While emotional critiques of Creative Ownership Disputes and Risk & AI Anxiety persisted, there was a notable rise in engagement with Futures & Speculation—a category associated with more optimistic expressions such as hope and curiosity about AI's potential in artistic and narrative domains. Meanwhile, institutional users remained largely concentrated on Policy & Regulation, although their involvement in Education & Integrity modestly increased, partly in response to public outrage over AI-assisted academic misconduct.

4.4.2 Factors underpinning topic fragmentation

Despite limited intersections, some thematic convergence can be observed in areas like Education & Integrity, where public pressure—such as viral hashtags related to AI ghostwriting—prompted institutional responses. However, the overall fragmentation in discourse can be attributed to three structural factors:

First, the scope and focus of ethical concerns differ substantially. Institutional users prioritize macro-level issues such as governance frameworks and compliance, while ordinary users emphasize micro-level concerns related to personal rights, authorship, and social justice. This divide is reinforced by their respective communication goals—stated consensus versus grassroots critique.

Second, discursive styles and emotional registers differ. Institutional users tend to adopt formal, policy-oriented language (“ethical compliance”, “cybersecurity enforcement”), whereas ordinary users engage through experiential, affective rhetoric (“AI is stealing my creativity”). These contrasting expressions result in parallel but rarely intersecting conversations, even when addressing similar issues such as intellectual property.

Third, platform governance and algorithmic visibility mediate exposure and amplification. During the policy roll-out phase, institutional content on data security and intellectual property benefited from platform promotion, while emotionally charged grassroots critiques were often downranked. In contrast, the release of DeepSeek-R1 catalyzed a viral surge in public engagement with sci-fi creation, prompting institutions to respond rhetorically but avoiding deeper structural questions around AI authorship.

In sum, the evolution of generative AI ethics discourse reflects a bifurcated dynamic: while institutional users remain anchored in state-sanctioned agendas, ordinary users

continue to drive emotional and participatory discourse. The Sankey diagram illustrates how this divergence manifests thematically and temporally, revealing the interplay between algorithmic governance and affective resistance in shaping China's digital public sphere.

5 Discussion

5.1 Theoretical implications: algorithmic publicness and affective counterpower in authoritarian contexts

5.1.1 From top-down visibility to emotionally differentiated counterpublics

Conventional agenda-setting models posit that institutional actors dominate public discourse through top-down control of issue salience. In authoritarian digital environments, this logic is extended through algorithmic infrastructures that amplify policy-aligned narratives while deprioritizing dissent. However, our sentiment analysis complicates this view by revealing that emotionally differentiated publics—notably negative affect around issues such as creative ownership (89%) and information authenticity (76%)—emerge as strategic counterforces to algorithmic silence.

Rather than being uniformly emotional or irrational, ordinary users exhibit selective negativity, often targeting institutional blind spots where accountability is lowest. Hashtags like #AISToleMyIdeas and #AICheating represent more than reactive outbursts; they exemplify affective counterpublics—user-driven formations that mobilize emotion as a mode of civic intervention. These publics may not overturn dominant discourse, but they succeed in disrupting visibility hierarchies, forcing institutional responses such as academic integrity reforms, even if symbolically.

This reframes emotional expression not as irrational noise but as embodied public reasoning (Papacharissi 2015), functioning under constraint to recalibrate the boundaries of sayable discourse.

5.1.2 Negotiated publicness and policy responsiveness under algorithmic constraint

Our findings support a shift from binary models of “control versus resistance” toward a more nuanced conceptualization of negotiated governance in platformed authoritarianism. Emotional content—particularly negatively valenced yet semantically coherent posts—operates within a logic of algorithmic constraint: visible enough to circulate within counterpublic spaces, but rarely algorithmically amplified to trend.

This condition produces what we term a controllable controversy: topics like AI plagiarism or misinformation are allowed limited discursive space, providing symbolic responsiveness while maintaining structural suppression. For instance, while Article 12 of the Interim Measures mandated AI labeling, it failed to address deeper platform-level opacity, reinforcing public skepticism and sustaining negative sentiment over time.

Our findings extend Han's (2022) notion of policy-driven publicness by demonstrating that emotion functions as both a bottom-up pressure mechanism and an institutional risk signal. Specifically, while positive sentiment is clustered around speculative domains (e.g., artistic creation), negative sentiment is concentrated in domains of normative contention—revealing where symbolic policy fails to address structural asymmetries.

In sum, affect under constraint does not merely reflect public mood; it performs governance signaling. It identifies algorithmic and institutional blind spots, mobilizes symbolic critique, and occasionally catalyzes policy correction—however, partial. Thus, we position affective publics as algorithmically legible yet structurally constrained political actors within an evolving ecosystem of negotiated publicness.

5.2 Policy implications: designing emotion-aware AI ethics communication

5.2.1 From policy promotion to emotion-anchored narratives

This study reveals clear affective variations across different AI ethics themes (see Fig. 1). Negative emotions—such as outrage (observed in 89% of posts on creative ownership) and anxiety (76% on information authenticity)—are concentrated in topics involving personal rights and social trust. Conversely, positive sentiment, particularly hope and curiosity, clusters around artistic and speculative themes, such as AI-assisted creation (68%).

These findings suggest that institutions should align their communication strategies with these emotional contours. For high-conflict topics, adopting *victim-centered narratives*—for example, “Digital artists speak out: How AI plagiarized my style”—can activate empathy and reinforce legitimacy. For more positive topics, forward-looking storytelling, such as “AI helped me overcome a creative block,” can foster optimism and participatory imagination. Such affective anchoring transforms policy messaging into emotionally resonant public discourse.

5.2.2 Emotion translators: mobilizing hybrid-affect influencers

Verified influencers (e.g., Gold V accounts) play a pivotal role in transmediating policy into emotionally engaging

formats. For instance, @PopScienceStar's viral post blending satire with existential anxiety gained 12,304 likes (as documented in Sect. 4.1, Table 1), demonstrating the power of hybrid affect in public engagement.

To enhance communication efficacy, authorities should co-produce content with such actors. Science communicators, for instance, can explain regulatory clauses through short-form humor videos—e.g., “Can you spot the deep-fake? When Messi sells you crypto.” Additionally, themes like AI's impact on labor can be framed through an anxiety–hope duality: “Is your job at risk? Here's how to pivot.” These emotion-driven strategies not only amplify reach but also build relational trust in otherwise opaque governance narratives.

5.2.3 Feedback channels for emotion migration and controversy management

Emotional discourse often spills over into alternative platforms when suppressed on mainstream ones (see Table 6). For example, after the hashtag #AICheating was suppressed on Weibo, users migrated to Bilibili and Zhihu, where related content gained millions of views and mobilized collective petitions. This cross-platform migration reveals the need for institutional feedback mechanisms that are both affect-aware and structurally responsive.

We recommend the implementation of dual-mode feedback channels. On domestic platforms like Weibo, policy response portals should allow users to submit emotion-tagged inputs—selecting from predefined emotions (see Table 2) alongside anonymized textual claims (e.g., “AI copied my design, no platform accountability!”). These submissions can be algorithmically clustered by emotion–theme patterns to inform policy prioritization.

Simultaneously, regulators should monitor emotionally charged spillovers on platforms such as Bilibili, Zhihu, and Douyin. Proactively addressing these alternative discursive arenas can help manage “controllable controversies” and preempt the escalation of unmanaged public grievances.

5.3 Limitations and future research

Despite its contributions, this study has several limitations. First, the sentiment classification was based on SnowNLP, a lexicon-based model calibrated through manual validation. While we ensured inter-coder reliability and provided clear thresholds, the model may still underperform in detecting nuanced or sarcastic expressions in Chinese social media, particularly under evolving platform norms. Future work could apply fine-tuned large language models (LLMs) or context-aware deep learning classifiers for enhanced accuracy.

Second, our theme extraction relies on unsupervised topic modeling (LDA) and post hoc manual refinement. Although coherence scores guided model selection and semantic overlap was minimized, LDA's bag-of-words limitation may affect topic distinctiveness. Structural Topic Modeling (STM) or transformer-based clustering could provide greater interpretability.

Third, while the study integrates affective patterns with governance dynamics, it is limited to textual data from Weibo. Excluding multimodal content (e.g., images, videos) and other platforms (e.g., Douyin, Zhihu) constrains the generalizability of affective publics. Cross-platform and multimodal analyses are needed to capture the full scope of emotion-driven discourse.

Finally, this study focuses on ethically themed discussions about generative AI. Broader applications of affective counterpublics—such as in environmental activism or public health crises—could further refine the theoretical model under diverse authoritarian settings.

6 Conclusion

This study demonstrates how policy-driven publicness and algorithmic curation jointly shape the discourse of AI ethics on Chinese social media. Drawing on over 20,000 Weibo posts across two regulatory milestones, we show that institutional actors largely reinforce compliance-oriented narratives, while ordinary users challenge these framings through emotionally saturated discourse around misinformation, authorship, and human well-being. Emotion—especially negative emotion—emerges not merely as a rhetorical device but as a grassroots mode of contestation within algorithmically managed publics.

6.1 Emotional agency in evolving AI discourse

Even under constrained digital governance, emotional agency persists and evolves in patterned ways. As shown in Fig. 1, negative affect—particularly outrage, anxiety, and satire—clusters around themes with immediate ethical implications such as creative ownership (89% negative sentiment) and information authenticity (76%). In contrast, hope and optimism are concentrated in more speculative domains such as artistic creation. These emotion-theme pairings reveal that user emotions are not arbitrary but issue-specific and temporally dynamic. Emotional discourse thus becomes a central modality through which publics sense, interpret, and act upon emerging AI technologies.

6.2 Theoretical contributions

This research contributes to ongoing debates in AI ethics and digital governance in three ways:

It extends agenda-setting theory to authoritarian contexts by illustrating how affectively mobilized counterpublics shape visibility hierarchies, despite algorithmic demotion.

It expands the concept of policy-driven publicness by showing that emotional resistance—particularly satire and anxiety—persists under platform constraints and can influence institutional discourse indirectly.

Methodologically, it introduces a structured approach to emotion-theme-user mapping using lexicon-based sentiment classification and visual analytics (e.g., radar-style emotion mapping), offering a scalable model for analyzing contentious publics in platform environments.

6.3 Toward emotionally responsive governance

Our findings point to practical strategies for making AI ethics governance more emotionally responsive and socially grounded. First, institutions should tailor narratives to specific emotional patterns: victim-centered storytelling can address outrage over plagiarism, while future-oriented framing can build hope around creative AI. Second, verified influencers—especially those blending satire with expert insight—can function as affect translators to bridge policy and public sentiment. Finally, feedback systems should allow emotion-tagged input and monitor cross-platform migration of emotional discourse. Rather than suppressing dissent, such measures can help institutions detect blind spots, preempt backlash, and craft more legitimate responses to the affective realities of AI governance.

7 Conflict of interest

The authors declare no competing interests.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00146-025-02437-0>.

Author contributions Bei Zhang conceived the study, conducted the research, performed the data analysis, and wrote the manuscript. All work was carried out solely by the author.

Funding Humanities and Social Sciences Research Project of Jiangxi Provincial Higher Education Institutions, China, XW22102.

Data availability No datasets were generated or analysed during the current study.

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